Vectorial Treatment of Certain Algebraic Theorems

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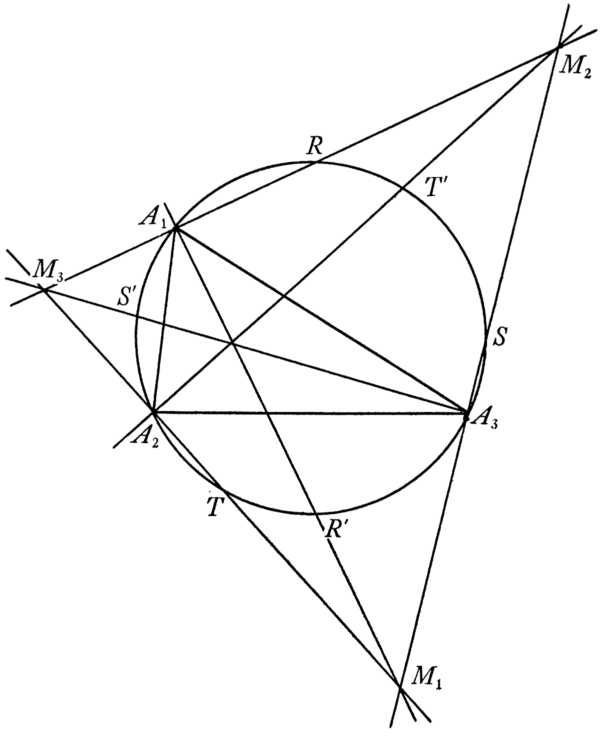

hypocycloid must coincide with the Simson line envelope of points on the circumcircle of triangle $M_1M_2M_3$ with respect to that triangle, each of the six lines mentioned above being such Simson lines.

VECTORIAL TREATMENT OF CERTAIN ALGEBRAIC THEOREMS

By T. C. ESTY, Amherst College

I. Consider the homogeneous linear equations

$$a_1x + b_1y + c_1z = 0, \tag{1}$$

$$a_2x + b_2y + c_2z = 0. \tag{2}$$

Let us define three vectors, α, β, ρ, as follows:

$$\alpha = a_1i + b_1j + c_1k, \quad \beta = a_2i + b_2j + c_2k, \quad \rho = xi + yj + zk, \tag{3}$$

where i, j, k form a right-handed system of mutually perpendicular unit vectors.

Equations (1), (2) may then be written as

$$\alpha \cdot \rho = 0, \tag{1'}$$

$$\beta \cdot \rho = 0. \tag{2'}$$

Equations (1′), (2′) are satisfied simultaneously by any vector ρ which is perpendicular to both α and β; that is, when ρ is given by

$$\rho = t\alpha \times \beta, \tag{4}$$

where t is any scalar.

Substituting for ρ, α, β their values from (3), expanding the vector product, and equating coefficients of i, of j, and of k, we obtain

$$x = t(b_1c_2 - b_2c_1), \quad y = t(c_1a_1 - c_2a_1), \quad z = t(a_1b_2 - a_2b_1), \tag{5}$$

showing that

$$\frac{x}{\begin{vmatrix} b_1 & c_1 \\ b_2 & c_2 \end{vmatrix}} = \frac{y}{\begin{vmatrix} c_1 & a_1 \\ c_2 & a_2 \end{vmatrix}} = \frac{z}{\begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix}}. \tag{6}$$

By putting $z=-1$ in (1) and (2) we obtain the system

$$\begin{aligned} a_1x + b_1y &= c_1, \\ a_2x + b_2y &= c_2, \end{aligned} \tag{7}$$

whose solution, as given by (6) when $z=-1$, is

$$x = \frac{\begin{vmatrix} c_1 & b_1 \\ c_2 & b_2 \end{vmatrix}}{\begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix}}, \qquad y = \frac{\begin{vmatrix} a_1 & c_1 \\ a_2 & c_2 \end{vmatrix}}{\begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix}}. \tag{8}$$

II. Consider the homogeneous linear equations,

$$a_1x + b_1y + c_1z = 0, \tag{1}$$

$$a_2x + b_2y + c_2z = 0, \tag{2}$$

$$a_3x + b_3y + c_3z = 0, \tag{3}$$

of which at least two, say (2) and (3), are independent. If we put

$$\begin{aligned} \alpha &= a_1i + b_1j + c_1k, \qquad & \beta &= a_2i + b_2j + c_2k, \\ \gamma &= a_3i + b_3j + c_3k, \qquad & \rho &= xi + yj + zk, \end{aligned} \tag{4}$$

then the given equations may be written in the form

$$\alpha\cdot\rho = 0, \tag{1′}$$

$$\beta\cdot\rho = 0, \tag{2′}$$

$$\gamma\cdot\rho = 0. \tag{3′}$$

We propose to find a necessary and sufficient condition that the last three equations have a common solution other than $\rho=0$; that is, that the given equations have a common solution other than $x=y=z=0$.

Let us first assume that there *is* such a value of ρ which satisfies $(1')$, $(2')$ and $(3')$. It follows that this ρ is perpendicular to each of the vectors α, β, γ, and hence that α, β, and γ are coplanar. This requires that

$$\alpha\cdot\beta\times\gamma = 0, \tag{5}$$

or

$$\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = 0. \tag{6}$$

This is a *necessary* condition.

Next, let us assume that (6), and hence (5), are fulfilled. Solving $(2')$ and $(3')$ for ρ in the form

$$\rho = t\beta\times\gamma \quad (\beta \text{ not parallel to } \gamma) \tag{7}$$

as in section I, we find by substitution in $(1')$,

$$\alpha\cdot\rho = t\alpha\cdot\beta\times\gamma,$$

and hence, by virtue of (5), $\alpha\cdot\rho$ vanishes identically. Thus $\rho=t\beta\times\gamma$, in which t is any scalar whatever, is a common solution of $(1')$, $(2')$ and $(3')$.

This proves that (5) is a *sufficient* condition that the equations have a common solution other than $\rho=0$.

It follows that (6) is a necessary and sufficient condition that equations (1), (2) and (3) have a common solution other than $x=y=z=0$.

One form of the solution may be found directly from (7), namely

$$x = t(b_2c_3 - b_3c_2), \quad y = t(c_2a_3 - c_3a_2), \quad z = t(a_2b_3 - a_3b_2), \tag{8}$$

where t is any scalar whatever.

By putting $z=-1$ in (1), (2), (3) we obtain the system

$$\begin{aligned} a_1x + b_1y &= c_1, \\ a_2x + b_2y &= c_2, \\ a_3x + b_3y &= c_3, \end{aligned} \tag{9}$$

and we see that (6) is a necessary condition that equations (9) have a common solution. If $a_2b_3-a_3b_2\neq 0$ the solution is unique; and it can be found from (8) by putting $z=-1$ and substituting the resulting value of t in the expressions for x and y.

III. Consider the system

$$\begin{aligned} a_1x + b_1y + c_1z &= d_1, \\ a_2x + b_2y + c_2z &= d_2, \\ a_3x + b_3y + c_3z &= d_3. \end{aligned} \tag{1}$$

We shall assume, as we may without loss of generality, that the signs have been so adjusted as to make the quantities d_1, d_2, d_3 all positive.

Let us now define three vectors α, β, γ as follows:

$$(2)\qquad \alpha = a_1 i + b_1 j + c_1 k,\quad \beta = a_2 i + b_2 j + c_2 k,\quad \gamma = a_3 i + b_3 j + c_3 k.$$

The equations (1) may be written in the equivalent form

$$(3)\qquad \alpha\cdot\rho = d_1,\quad \beta\cdot\rho = d_2,\quad \gamma\cdot\rho = d_3.$$

If we regard α, β, γ, ρ as position vectors with a common origin at 0, these equations represent three planes which are perpendicular to α, β, γ, respectively, and at distances from 0 given by d_1/a, d_2/b, d_3/c, where a, b, c are the lengths of α, β, γ.

We now introduce the familiar formula

$$(4)\qquad \rho(\alpha\cdot\beta\times\gamma) = \beta\times\gamma(\alpha\cdot\rho) + \gamma\times\alpha(\beta\cdot\rho) + \alpha\times\beta(\gamma\cdot\rho)$$

in which α, β, γ are supposed to be any three non-coplanar vectors. On the supposition, then, that the vectors α, β, γ in (2) are non-coplanar, that is, that $\alpha\cdot\beta\times\gamma\neq 0$, we have from (3) and (4)

$$(5)\qquad \rho = \frac{d_1\beta\times\gamma + d_2\gamma\times\alpha + d_3\alpha\times\beta}{\alpha\cdot\beta\times\gamma}.$$

This gives the position vector of the point common to planes (3). Replacing ρ by $xi+yj+zk$ and α, β, γ by their values in (2) we get

$$(6)\quad xi + yj + zk = \frac{d_1\begin{vmatrix} i & j & k \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} + d_2\begin{vmatrix} i & j & k \\ a_3 & b_3 & c_3 \\ a_1 & b_1 & c_1 \end{vmatrix} + d_3\begin{vmatrix} i & j & k \\ a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \end{vmatrix}}{\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}}.$$

Expanding the determinants in the numerator it is found that the i-terms are

$$[d_1(b_2c_3 - b_3c_2) + d_2(b_3c_1 - b_1c_3) + d_3(b_1c_2 - b_2c_1)]i = \begin{vmatrix} d_1 & b_1 & c_1 \\ d_2 & b_2 & c_2 \\ d_3 & b_3 & c_3 \end{vmatrix} i$$

with similar results for the j- and k-terms.

Equating coefficients of like vectors in (6) we obtain

$$(7)\qquad x=\frac{\begin{vmatrix} d_1 & b_1 & c_1\\ d_2 & b_2 & c_2\\ d_3 & b_3 & c_3\end{vmatrix}}{\begin{vmatrix} a_1 & b_1 & c_1\\ a_2 & b_2 & c_2\\ a_3 & b_3 & c_3\end{vmatrix}},\quad y=\frac{\begin{vmatrix} a_1 & d_1 & c_1\\ a_2 & d_2 & c_2\\ a_3 & d_3 & c_3\end{vmatrix}}{\begin{vmatrix} a_1 & b_1 & c_1\\ a_2 & b_2 & c_2\\ a_3 & b_3 & c_3\end{vmatrix}},\quad z=\frac{\begin{vmatrix} a_1 & b_1 & d_1\\ a_2 & b_2 & d_2\\ a_3 & b_3 & d_3\end{vmatrix}}{\begin{vmatrix} a_1 & b_1 & c_1\\ a_2 & b_2 & c_2\\ a_3 & b_3 & c_3\end{vmatrix}}.$$

Just as (5) furnishes a single value of ρ when $\alpha\cdot\beta\times\gamma\neq 0$, so will equations (1) have a single solution when

$$(8)\qquad \begin{vmatrix} a_1 & b_1 & c_1\\ a_2 & b_2 & c_2\\ a_3 & b_3 & c_3\end{vmatrix}\neq 0.$$

As a numerical example of the foregoing we shall solve the equations

$$\begin{aligned} x+6y-5z&=23,\\ -3x+8y-4z&=1,\\ 7x-10y+10z&=0.\end{aligned}$$

We put

$$\alpha=i+6j-5k,\qquad \beta=-3i+8j-4k,\qquad \gamma=7i-10j+10k$$

and find

$$\beta\times\gamma=40i+2j-26k,\ \gamma\times\alpha=-10i+45j+52k,\ \alpha\cdot\beta\times\gamma=182.$$

We also have $d_1=23$, $d_2=1$, $d_3=0$.

Substituting these values in (5) we get

$$\rho=xi+yj+zk=\frac{23(40i+2j-26k)+(-10i+45j+52k)}{182}=5i+\tfrac{1}{2}j-3k,$$

Whence $x=5$, $y=\frac{1}{2}$, $z=-3$.

Special cases which can arise under the supposition that $\alpha\cdot\beta\times\gamma$ *is* equal to zero, that is when

$$(9)\qquad \rho(0)=d_1\beta\times\gamma+d_2\gamma\times\alpha+d_3\alpha\times\beta,$$

may be considered under the following heads:

(1) When α, β, and γ are coplanar, without further restrictions.

Here there are two possibilities; (a) when the right side of (9) is *not* equal to zero, and (b) when it *is* equal to zero.

Under (a), there is no solution. In fact equations (3) show that the three planes are perpendicular to the plane of α, β, γ, and these planes intersect by pairs in three parallel lines whose direction is that of $\alpha\times\beta$, Under (*b*),

$$d_1\beta \times \gamma + d_2\gamma \times \alpha + d_3\alpha \times \beta = 0. \tag{10}$$

To interpret this result we note that γ is in the plane of α and β; hence we may put

$$\gamma = l\alpha - m\beta. \tag{11}$$

Then $d_1\beta\times\gamma = ld_1\beta\times\alpha = -ld_1\alpha\times\beta$, $d_2\gamma\times\alpha = -md_2\beta\times\alpha = md_2\alpha\times\beta$ and (10) becomes $(-ld_1+md_2+d_3)\alpha\times\beta=0$, or since $\alpha\times\beta\neq 0$,

$$d_3 = ld_1 - md_2. \tag{12}$$

Substituting from (11) and (12) in the equation $\gamma\cdot\rho=d_3$, the equation of that plane becomes

$$l(\alpha\cdot\rho - d_1) - m(\beta\cdot\rho - d_2) = 0, \tag{13}$$

which is the equation of a plane through the intersection of the two planes $\alpha\cdot\rho=d_1$ and $\beta\cdot\rho=d_2$. Hence planes (3) *meet in one line*, which is parallel to $\alpha\times\beta$, and every point of this line is a solution of equations (3).

The corresponding values of x, y, z are therefore solutions of equations (1), and we say that these equations have an infinite number of solutions. To find the relations between the constants in (1) which exist in this case, we substitute the values of α, β, γ in (10) and expand the vector products. Then collecting the i-, j-, and k-terms separately, we get

$$\begin{vmatrix} d_1 & b_1 & c_1 \\ d_2 & b_2 & c_2 \\ d_3 & b_3 & c_3 \end{vmatrix} i + \begin{vmatrix} a_1 & d_1 & c_1 \\ a_2 & d_2 & c_2 \\ a_3 & d_3 & c_3 \end{vmatrix} j + \begin{vmatrix} a_1 & b_1 & d_1 \\ a_2 & b_2 & d_2 \\ a_3 & b_3 & d_3 \end{vmatrix} k = 0, \tag{14}$$

which requires that the three determinants vanish separately. In addition to these conditions we have, of course, since $\alpha\cdot\beta\times\gamma=0$:

$$\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = 0. \tag{15}$$

It is interesting to note that when equations (1) represent three planes with a common line of intersection, the equations are not independent, for by (11)

$$a_3 = la_1 - ma_2,\quad b_3 = lb_1 - mb_2,\quad c_3 = lc_1 - mc_2, \tag{11$'$}$$

and by (12)

$$d_3 = ld_1 - md_2. \tag{12$'$}$$

These relations show that the third of the given equations can be obtained by multiplying the first by a number l and subtracting from the result the second multiplied by a number m. The process of finding the values of l and m will be illustrated in the following example.

Solve

$$\begin{aligned} 2x - 3y + 4z &= 5, \\ 3x + y - 2z &= 1, \\ -5x - 9y + 14z &= 7. \end{aligned}$$

Here $\alpha = 2i - 3j + 4k$, $\beta = 3i + j - 2k$, $\gamma = -5i - 9j + 14k$, and

$$\alpha \cdot \rho = 5, \quad \beta \cdot \rho = 1, \quad \gamma \cdot \rho = 7. \tag{3'}$$

Also $\alpha \cdot \beta \times \gamma = 0$, $\beta \times \gamma = -4i - 32j - 22k$, $\gamma \times \alpha = 6i + 48j + 33k$, $\alpha \times \beta = 2i + 16j + 11k$, so that

$$d_1\beta \times \gamma + d_2\gamma \times \alpha + d_3\alpha \times \beta = 5\beta \times \gamma + \gamma \times \alpha + 7\alpha \times \beta = 0.$$

Since no two of the vectors α, β, γ are parallel and since $d_1\beta \times \gamma + d_2\gamma \times \alpha + d_3\alpha \times \beta = 0$, the planes (3′) must meet in one line, and the given equations have an infinite number of solutions.

From the first of equations (11′) we have $2l - 3m = -5$, and from (12′) $5l - m = 7$. Solving these two equations, we find $l = 2$, $m = 3$. In fact, if we multiply the first of the given equations by 2 and subtract three times the second, we obtain the third.

If the equations of the line of intersection are desired, we may proceed as follows.

Let us first find the point in which the line pierces the plane of i and j. This point of the line may be regarded as the intersection of the three planes $\alpha \cdot \rho = 5$, $\beta \cdot \rho = 1$ and $k \cdot \rho = 0$. Denoting the vector of their common point by σ, and using the formula

$$\rho = \frac{d_1\beta \times \gamma + d_2\gamma \times \alpha + d_3\alpha \times \beta}{\alpha \cdot \beta \times \gamma}.$$

we have

$$\sigma = \frac{5\beta \times k + k \times \alpha}{\alpha \cdot \beta \times k} = \frac{8i - 13j}{11}.$$

Since the line is parallel to $\alpha \times \beta$, its equation will have the form $\rho = \sigma + t\alpha \times \beta$, where t is a variable scalar. Thus

$$xi + yj + zk = \frac{8}{11}i - \frac{13}{11}j + t(2i + 16j + 11k)$$

whence

$$\frac{x - \frac{8}{11}}{2} = \frac{y + \frac{13}{11}}{16} = \frac{z}{11}.$$

(2) When α, β, γ are coplanar, as before, but in addition, two of them, β and γ, (say) are parallel.

Here $\beta\times\gamma=0$, and from (9)

$$\rho(0) = d_2\gamma \times \alpha + d_3\alpha \times \beta. \tag{16}$$

We have two possible cases; (a) when the right side is *not* equal to zero, and (b) when the right side *is* equal to zero.

Under (a) there is no solution. Two of the planes are parallel and are cut by the third plane in two lines which are parallel to $\alpha\times\beta$. Furthermore, since $\beta\times\gamma=0$, we have

$$\frac{a_2}{a_3} = \frac{b_2}{b_3} = \frac{c_2}{c_3}. \tag{17}$$

Under (b),

$$d_2\gamma\times\alpha+d_3\alpha\times\beta=0 \tag{18}$$

To interpret this, we first note that β and γ are supposed to be parallel. Now if they agree in sense, the vectors $\gamma\times\alpha$ and $\alpha\times\beta$ differ in sense. On the other hand, if β and γ differ in sense, then $\gamma\times\alpha$ and $\alpha\times\beta$ agree in sense. Under the latter supposition (18) is impossible, since d_2 and d_3 are positive. But under the former supposition, (18) becomes

$$d_2c - d_3b = 0, \text{ or } d_2/b = d_3/c. \tag{19}$$

Now d_2/b and d_3/c are the distances from 0 of the planes $\beta\cdot\rho=d_2$ and $\gamma\cdot\rho=d_3$, respectively, and since these distances are equal, the two planes coincide. All points of the line in which these coincident planes are cut by the third plane are solutions of (3), and hence equations (1) have in this case an infinite number of solutions.

It is easy to determine the conditions under which equations (1) present this case, for, since β is parallel to γ, and of the same sense, we may write

$$\beta = n\gamma, \text{ whence } n = b/c = d_2/d_3. \tag{20}$$

Also

$$a_2i + b_2j + c_2k = n(a_3i + b_3j + c_3k), \tag{21}$$

whence by (20) and (21)

$$a_2/a_3 = b_2/b_3 = c_2/c_3 = d_2/d_3. \tag{22}$$

(3) When all three vectors α, β, γ are parallel. Then, separately,

$$\beta \times \gamma = 0, \ \gamma \times \alpha = 0, \ \alpha \times \beta = 0. \tag{23}$$

In this case all of the planes (3) are parallel and in general there is no solution. We see from (23) that this situation arises when

$$a_2/a_3 = b_2/b_3 = c_2/c_3, \ a_3/a_1 = b_3/b_1 = c_3/c_1, \text{ and } a_1/a_2 = b_1/b_2 = c_1/c_2. \tag{24}$$

If, however, α, β, and γ agree in sense, and, in addition, $d_1/a=d_2/b=d_3/c$, then all three planes coincide, and, as in $2(b)$ the groups (24) of equal ratios include as a fourth member d_2/d_3, d_3/d_1, and d_1/d_2 respectively.

IV. To express the square of a determinant of the third order as a determinant of the same order.

Consider

$$(1)\qquad \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}^2.$$

Introducing the vectors

$$\alpha = a_1i + a_2j + a_3k,\ \beta = b_1i + b_2j + b_3k,\ \gamma = c_1i + c_2j + c_3k.$$

we may write (1) as

$$\begin{aligned}(\alpha\cdot\beta\times\gamma)(\alpha\cdot\beta\times\gamma) &= \alpha\cdot\beta\times[\gamma(\alpha\cdot\beta\times\gamma)],\\ &= \alpha\cdot\beta\times[\gamma(\alpha\cdot\beta\times\gamma)-\alpha(\gamma\cdot\beta\times\gamma)],\end{aligned}$$

since $\gamma\cdot\beta\times\gamma=0$. Then

$$(2)\qquad (\alpha\cdot\beta\times\gamma)(\alpha\cdot\beta\times\gamma) = \alpha\cdot\beta\times[(\beta\times\gamma)\times\gamma\times\alpha],$$

by the formula for a vector triple product; or

$$\begin{aligned}(\alpha\cdot\beta\times\gamma)(\alpha\cdot\beta\times\gamma) &= \alpha\times\beta\cdot[(\beta\times\gamma)\times(\gamma\times\alpha)]\\ &= (\alpha\times\beta)\cdot(\beta\times\gamma)\times(\gamma\times\alpha)\end{aligned}$$

which is equivalent to

$$(3)\qquad \begin{vmatrix} a_2b_3-a_3b_2 & a_3b_1-a_1b_3 & a_1b_2-a_2b_1 \\ b_2c_3-b_3c_2 & b_3c_1-b_1c_3 & b_1c_2-b_2c_1 \\ c_2a_3-c_3a_2 & c_3a_1-c_1a_3 & c_1a_2-c_2a_1 \end{vmatrix}.$$

V. To express the product of two determinants of the third order as a single determinant of the same order.

Consider the product

$$(1)\qquad \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} \times \begin{vmatrix} l_1 & l_2 & l_3 \\ m_1 & m_2 & m_3 \\ n_1 & n_2 & n_3 \end{vmatrix}.$$

Introduce the vectors

$$(2)\qquad \begin{aligned} \alpha &= a_1i + a_2j + a_3k, & \beta &= b_1i + b_2j + b_3k, & \gamma &= c_1i + c_2j + c_3k,\\ \lambda &= l_1i + l_2j + l_3k, & \mu &= m_1i + m_2j + m_3k, & \nu &= n_1i + n_2j + n_3k. \end{aligned}$$

Then the product (1) may be expressed as

$$
\begin{aligned}
(3)\quad & (\alpha\cdot\beta\times\gamma)(\lambda\cdot\mu\times\nu)\\
&= \alpha\cdot\beta\times\left[\gamma(\lambda\cdot\mu\times\nu)\right]\\
&= \alpha\cdot\beta\times\left[\mu\times\nu(\gamma\cdot\lambda)+\nu\times\lambda(\gamma\cdot\mu)+\lambda\times\mu(\gamma\cdot\nu)\right]\\
&= \alpha\cdot\beta\times\mu\times\nu(\gamma\cdot\lambda)+\alpha\cdot\beta\times\nu\times\lambda(\gamma\cdot\mu)+\alpha\cdot\beta\times\lambda\times\mu(\gamma\cdot\nu)\\
&= \alpha\cdot\left[\mu(\beta\cdot\nu)-\nu(\beta\cdot\mu)\right](\gamma\cdot\lambda)+\alpha\cdot\left[\nu(\beta\cdot\lambda)-\lambda(\beta\cdot\nu)\right](\gamma\cdot\mu)\\
&\quad+\alpha\cdot\left[\lambda(\beta\cdot\mu)-\mu(\beta\cdot\lambda)\right](\gamma\cdot\nu)\\
&= (\alpha\cdot\mu)(\beta\cdot\nu)(\gamma\cdot\lambda)-(\alpha\cdot\nu)(\beta\cdot\mu)(\gamma\cdot\lambda)+(\alpha\cdot\nu)(\beta\cdot\lambda)(\gamma\cdot\mu)-(\alpha\cdot\lambda)(\beta\cdot\nu)(\gamma\cdot\mu)\\
&\quad+(\alpha\cdot\lambda)(\beta\cdot\mu)(\gamma\cdot\nu)-(\alpha\cdot\mu)(\beta\cdot\lambda)(\gamma\cdot\nu).
\end{aligned}
$$

Or

$$
(4)\qquad (\alpha\cdot\beta\times\gamma)(\lambda\cdot\mu\times\nu)=\begin{vmatrix}\alpha\cdot\lambda & \alpha\cdot\mu & \alpha\cdot\nu\\ \beta\cdot\lambda & \beta\cdot\mu & \beta\cdot\nu\\ \gamma\cdot\lambda & \gamma\cdot\mu & \gamma\cdot\nu\end{vmatrix}
$$

$$
=\begin{vmatrix} a_1l_1+a_2l_2+a_3l_3 & a_1m_1+a_2m_2+a_3m_3 & a_1n_1+a_2n_2+a_3n_3\\ b_1l_1+b_2l_2+b_3l_3 & b_1m_1+b_2m_2+b_3m_3 & b_1n_1+b_2n_2+b_3n_3\\ c_1l_1+c_2l_2+c_3l_3 & c_1m_1+c_2m_2+c_3m_3 & c_1n_1+c_2n_2+c_3n_3\end{vmatrix}.
$$

QUESTIONS AND DISCUSSIONS

EDITED by R. E. GILMAN, Brown University, Providence, Rhode Island.

The Department of Questions and Discussions in the Monthly is open to all forms of activity in collegiate mathematics, including the teaching of mathematics, except for specific problems, especially new problems, which are reserved for the department of Problems and Solutions.

ON THE SOLUTIONS OF THE EQUATION $\binom{x+1}{y}=\binom{x}{y+1}$.

By W. RANDOLPH CHURCH, Hamden, Conn.

In the June–July issue of this Monthly, F. Underwood[1] solved the problem of determining all integral values of x and y which satisfy the equation

$$
(1)\qquad \binom{x}{y+1}=\binom{x+1}{y}.
$$

It is the purpose of the present note to bring out certain aspects of the subject which may be of interest. In particular, a different solution will be outlined in which it will not be necessary to distinguish different cases, and which will be

[1] This Monthly, vol. 38 (1931), pp. 351–354. The problem was proposed by Norman Anning, ibid., vol. 37 (1930), p. 508.